Ashwini Kumar

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Education

Indian Institute Of Technology (ISM) — Dhanbad, India
B.Tech in Computer Science and Engineering (CGPA: 9.04) — *2016–2020*

Experience

Samsung Research Institute — Bangalore, India
Lead Software Engineer — *Jan 2021 – Present*

- **3GPP Standardization Study**: Led the study items for AI based applications in **5G Advanced and 6G systems**. Contributed to **over 10 Technical-Documents** in 3GPP. Built multiple robust AI models to perform **CSI compression, CSI prediction, beam management, and UE position enhancement**. The AI models had minimal overhead and specification impact, improving performance by 20% over conventional systems across various use-cases.
- **UE side optimization**: Designed an algorithm to efficiently download AI models from a server to a User Equipment(UE), which helped in reducing the overhead by 33%. Developed **Federated learning** algorithm for model updates at the UE side, which reduced latency by 25%.
- **Network side optimization**: Designed an algorithm to assign a **Mean Network Opinion Score(MNOS)** to the network, which helps the UE intelligently decide which network to latch onto. Designed an algorithm to ensure that UE avoids Radio Link Failure by assessing the network conditions; this idea has been taken up for **commercialization.**
- **Smart-CSI**: Designed an AI-based CSI prediction system to ensure that the most accurate CSI measurements are used for throughput scheduling. This helped in achieving 15% gains over the no-prediction scenario. This idea is the **World's first** use of AI in modem. The idea has been **commercialized** in USA.
- **LLMs for Wireless communication**: Designed and implemented a specialized LLM for Wireless communication, capable of reasoning over large channel datasets to recommend parameter tuning for resource allocation, scheduling, and beamforming.
- **Human Activity Recognition**: Designed an ensemble ML model using CNN to perform human activity recognition using images generated by a radar. This solution won the **first prize** at the ITU AI/ML Challenge in 2021.
- **AI-based Digital Pre-distorter**: Designed and implemented an ML model to perform digital pre-distortion in amplifiers to ensure that they remain well within their operating range.
- **Team & Tech Leadership**: Led and mentored a team of engineers, driving feature development and operations through rigorous design reviews and architectural enforcement.
- **Software Certification** : Aced the Software Certification test(Advanced) Organized by Samsung headquarters(South Korea).

Postman — Bangalore, India
Software Engineer Intern — *August 2020 - September 2020*

- **Role** : Developed a customised Monaco text editor used for receiving inputs in the Postman application, which helped in providing a seamless user experience.

Papers and Patents

- **Patents:** Inventor on over 10 granted and filed global patents in AI-driven wireless communication, including innovations in CSI compression, CSI prediction, AI model downloading, and intelligent resource management.
- **Publications:**Co-authored 4 research papers presented at leading international conferences (IEEE, Globecom, ICC) on applying AI to next-generation communication systems.

Awards

- 1st Prize – ITU AI/ML Challenge (2021)
- AI Expert Certification – Workera
- Samsung Excellence Award – Commercialization of Smart-CSI
- Samsung Spot Award – Contributions to 3GPP Rel-18/19/20

Technical Skills

- **AI/ML Expertise:** Deep Learning (CNN, RNN, Transformers), Generative AI, Large Language Models (LLMs), Prompt Engineering, Federated Learning, Reinforcement Learning, Ensemble Learning
- **Languages:** Python, C/C++, Java, JavaScript, TypeScript, Node.js
- **Tools and Libraries:** MATLAB, TensorFlow, PyTorch, Keras, Scikit-learn
- **Concepts:** Distributed Systems, Generative Artificial Intelligence, Microservices, Asynchronous Processing, CI/CD, Security Compliance